

Large-Scale Stellar Bars and Nuclear Activity in Narrow-Line Seyfert 1 Galaxies

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ABSTRACT

We present a statistical comparison between barred and unbarred narrow-line Seyfert 1 galaxies (NLS1s) to investigate the role of large-scale stellar bars in fueling AGN with low-mass black holes. From a parent sample of 492 NLS1s selected from SDSS and Pan-STARRS1 (PS1), we construct a PS1-resolved matched sample of 101 barred and 101 unbarred NLS1s, matched in redshift, inclination, and image quality. Unbarred NLS1s are modestly but consistently brighter than barred NLS1s in several optical AGN luminosity indicators, including $\lambda L_{\lambda}(5100 \text{ \AA})$, broad $H\alpha$, broad $H\beta$, narrow $H\alpha$, narrow $H\beta$, and $[\text{O III}] \lambda 5007$, with typical offsets of $\sim 0.1\text{--}0.25$ dex; the broad $H\alpha$ and narrow-line offsets are statistically significant, while the continuum and broad $H\beta$ offsets are marginal. Monte Carlo resampling and a Bayesian model support the same direction of offset. In contrast, the main narrow-line diagnostics, including $[\text{O III}]/H\beta$, O3N2 , and the $[\text{S II}]$ -based electron density, are statistically consistent between the subsamples, although barred NLS1s show a higher N2 ratio. Within the barred subsample, the bar strength parameter e_{max} shows a mild anti-correlation with the optical continuum and broad Balmer-line luminosities. These results indicate that large-scale stellar bars are not associated with enhanced instantaneous optical AGN luminosity. Instead, barred systems may experience more regulated or time-variable inner gas delivery, building circumnuclear gas reservoirs without necessarily increasing accretion power observed at the epoch traced by optical AGN emission.

Keywords: Galaxy bars (2364) — Seyfert galaxies (1447) — Active galactic nuclei (16) — Galaxy structure (622) — Galaxy evolution (594)

1. INTRODUCTION

Active galactic nuclei (AGNs) may play a central role in the coevolution of galaxies and their central supermassive black holes (Kormendy & Ho 2013; Heckman & Best 2014). A major open question in AGN research is how material is transported from galactic scales into the nuclear regions, where it can fuel accretion onto the central black hole (Combes et al. 2019; Audibert et al. 2019). Two primary mechanisms have been proposed: large-scale stellar bars (Shlosman et al. 1989; Ho et al. 1997) and galaxy mergers (Sanders et al. 1988; Barnes & Hernquist 1991). In particular, stellar bars are thought to efficiently redistribute angular momentum and funnel gas inward from kiloparsec scales to the central few hundred parsecs, where it may trigger both nuclear star

formation and AGN activity (Ho et al. 1997; Martin & Friedli 1997; Sakamoto et al. 1999; Hopkins & Quataert 2010; Ellison et al. 2011).

The influence of large-scale bars on AGNs, however, remains controversial. Some studies have found that bar fractions are similar in AGN hosts and inactive galaxies (Ho et al. 1997; Hunt & Malkan 1999; Cisternas et al. 2015; Zee et al. 2023), whereas others reported that AGNs show enhanced bar fractions (Knapen et al. 2000; Laurikainen et al. 2004; Silva-Lima et al. 2022). From the perspective of AGN strength, Alonso et al. (2013, 2014, 2018) and Marels et al. (2025) showed that barred AGNs tend to have higher $[\text{O III}]$ luminosities and younger stellar populations than their unbarred counterparts. In contrast, Cisternas et al. (2013); Cisternas et al. (2015) and Galloway et al. (2015) reported no significant difference in nuclear activities, and Goulding et al. (2017) concluded that bars do not significantly affect black hole accretion rates in nearby galaxies ($z < 0.15$).

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Beyond their presence, the strength of bars may also play a role. While some studies have linked strong bars to enhanced star formation (Aguerri 1999), others have associated them with reduced star formation efficiency, older stellar populations, redder colors, and higher metallicities (Vera et al. 2016; Zee et al. 2023). These results suggest that both the presence and strength of bars may influence AGN fueling and host galaxy evolution, though the overall picture remains an area of ongoing research.

AGNs are believed to be powered by accretion disks surrounding supermassive black holes, with different types exhibiting distinct emission-line properties (Khachikian & Weedman 1974; Osterbrock 1981). Narrow-line Seyfert 1 galaxies (NLS1s) are a unique class of AGNs (Osterbrock & Pogge 1985). Unlike typical Seyfert 1 galaxies (broad-line Seyfert 1; BLS1s), which display broad $H\beta$ emission lines with FWHM greater than 2000 km/s, NLS1s are characterized by relatively narrower broad permitted emission lines with FWHM less than 2000 km/s (Goodrich 1989). They stand out due to their lower supermassive black hole (SMBH) masses, typically ranging from $10^6 - 10^8 M_\odot$, which is significantly less than those found in other Seyfert galaxies and many quasars (Peterson et al. 2000; Zhou et al. 2006). These galaxies also exhibit high Eddington ratios, indicating efficient and rapid black hole growth (Mathur 2000). The combination of large-scale environments, lower black hole masses and high Eddington ratios suggests that NLS1s may represent an early phase in AGN evolution, where black holes are actively accreting matter before evolving into more typical Seyfert galaxies or quasars (Järvelä et al. 2017; Kim & Ho 2019; Mathur 2000; Winkel et al. 2022).

Previous studies have predominantly examined Seyfert 1, Seyfert 2 galaxies, and LINERs, which typically feature high black hole masses and lower accretion rates, indicating they are in more advanced evolutionary stages. However, the influence of large-scale bars on AGNs may not be prominently observed or strongly impactful at these stages. Focusing on large-scale bars of NLS1s offers a promising way to deepen our understanding of how AGNs form and evolve, potentially shedding light on the role of large-scale bars in this process.

Several studies have indeed suggested a strong connection between bars and NLS1s. Kim et al. (2017) reported that the bar fraction among NLS1s (48%) is significantly higher than that among BLS1s (22%), consistent with earlier results (Crenshaw et al. 2003; Deo et al. 2006; Ohta et al. 2007; Orban de Xivry et al. 2011). However, these works were limited by small sample sizes, often fewer than 50 NLS1s, leaving the statistical signif-

icance uncertain. To establish a clearer picture, larger and more systematic studies are required, combining reliable bar classifications with multi-wavelength diagnostics of AGN activity.

Motivated by (1) the possibility that NLS1s represent an early phase of AGN evolution, (2) the scarcity of large-sample studies linking host galaxy morphology with the properties of NLS1s, and (3) the need to investigate the role of bars in AGNs at early stages, this work presents a statistical comparison between barred and unbarred NLS1 galaxies. Using Pan-STARRS (PS1; Chambers et al. 2016; Flewelling et al. 2020; Magnier et al. 2020a,b,c; Waters et al. 2020) imaging, we classify bar structures in NLS1 hosts, while Sloan Digital Sky Survey (SDSS) (York et al. 2000) spectra provide optical emission-line diagnostics. By combining host morphology and spectroscopic measurements, this study aims to clarify the role of bars in fueling AGNs and shaping the evolutionary pathways of NLS1 galaxies.

The paper is organized as follows. In Section 2, we describe the sample selection, bar identification, and matched subsample pairing. Section 3 presents the statistical analysis of optical spectra for barred and unbarred NLS1s. Section 4 discusses the implications of our results for AGN fueling. Finally, Section 5 summarizes our main conclusions.

2. DATA AND METHOD

2.1. Imaging Data

The PS1 all-sky survey provides extensive imaging datasets in the g , r , i , z , and y bands for morphological studies (Chambers et al. 2016; Flewelling et al. 2020). Among these, g -band images are more affected by dust extinction, while the stacked y -band data are relatively shallow, with a mean 5σ point-source limiting sensitivity of ~ 21.4 mag (Chambers et al. 2016). For this work, we therefore employ PS1 r -band stacked images for the morphological classification of NLS1 host galaxies. In the matched sample, these data have a typical effective PSF size of $\sim 1.1''$ (Table 1), compared with the median r -band seeing of $\sim 1.4''$ in SDSS imaging (Lupton et al. 2001). The PS1 stacked r -band data also reach a mean 5σ point-source limiting sensitivity of ~ 23.2 mag (Chambers et al. 2016). This combination provides uniform photometric coverage for the SDSS spectroscopic NLS1 sample.

A comprehensive catalog of NLS1s from the SDSS Data Release 17 (DR17; Abdurro'uf et al. 2022) was presented by Paliya et al. (2024), containing 22,656 objects. This sample is nearly two times larger than previous NLS1 compilations (Rakshit et al. 2017). Instead of using the $H\beta$ FWHM threshold of 2200 km/s without con-

sidering the uncertainties (Rakshit et al. 2017), Paliya et al. (2024) followed the traditional definition of NLS1 galaxies, e.g., the FWHM of the broad H β emission line within measured uncertainties is smaller than 2000 km/s (Osterbrock & Pogge 1985). The catalog provides key AGN spectral properties, including nuclear monochromatic luminosity at 5100 Å, luminosities of broad and narrow lines, velocity information, and accretion properties. This dataset is therefore well suited to extend the NLS1 sample for the present study.

We cross-matched the catalog of Paliya et al. (2024) with the PS1 survey and retrieved *r*-band stacked image cutouts through the STScI PS1 Image Cutout Service¹. At a representative redshift of $z = 0.1$, this corresponds to a physical resolution of ~ 1.8 kpc/arcsec, which is sufficient to resolve typical large-scale bars with sizes of 3–5 kpc. In total, 492 NLS1 galaxies with redshifts < 0.1 were identified within the PS1 footprint. These objects constitute the parent sample for our bar identification analysis.

2.2. Bar Identification

We identified bar structures using the ELLIPSE task (Jedrzejewski 1987) in IRAF (Tody 1986, 1993), which has been widely employed to characterize bars by measuring ellipticity (e), position angle (PA), and surface brightness as a function of radius. Bars are characterized by isophotes that remain elongated with nearly constant orientation across the bar region, followed by a marked decrease in ellipticity and a significant change in PA at the bar end. Accordingly, we adopted the bar identification criteria of Marinova & Jogee (2007): (a) the maximum ellipticity ($e_{\max}$) of a bar must exceed 0.25; (b) the PA should remain stable within 10° across the bar region; and (c) at the bar end, the ellipticity should decrease by > 0.1 accompanied by a PA change of $> 10^\circ$. We masked bright stars on the images and then fitted ellipse to each image out to where the galaxy isophotes reach the sky level. An example is shown in Figure 1.

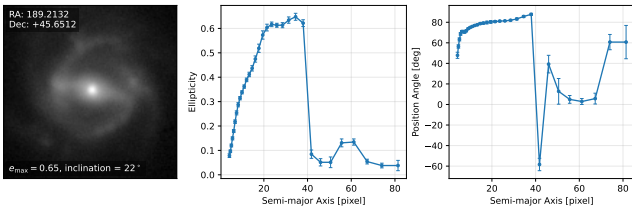


Figure 1. An example of bar identification. Left: Galaxy image; Middle: ellipticity (e) vs. semi-major axis; Right: PA vs. semi-major axis

¹ <https://ps1images.stsci.edu/cgi-bin/ps1cutouts>

Bar structures could be misidentified due to projection effects of high inclination angles. We estimated the inclination angles using the disk axis ratio, adopting the standard approximation: $\cos^2 i = \frac{(b/a)^2 - q_0^2}{1 - q_0^2}$, b/a is the disk axis ratio, $q_0 = 0.2$ is the assumed intrinsic thickness of spiral galaxies (Holmberg 1958; Padilla & Strauss 2008). The disk axis ratio (b/a) was determined from the ellipticity measured in the outer isophotes with semi-major axes $sma \geq 0.6 sma_{\max}$, where $sma_{\max}$ is the largest fitted semi-major axis. Galaxies with inclination angles $i > 60^\circ$ were excluded, as bar identification is more reliable in moderately inclined disk galaxies (e.g., Marinova & Jogee 2007; Sheth et al. 2008).

To ensure that bar classifications were based on sufficiently resolved galaxy structures, we required $sma_{\max}$ to be larger than four times the effective PS1 resolution, $FWHM_{\text{eff}}$:

$$\frac{sma_{\max}}{FWHM_{\text{eff}}} > 4,$$

where the $FWHM_{\text{eff}}$ for each image is the geometric mean of the major and minor FWHM values in the PS1 StackObjectView table (Flewelling et al. 2020):

$$FWHM_{\text{eff}} = \sqrt{FWHM_{\text{major}} \times FWHM_{\text{minor}}}.$$

This provides a single representative value of the PSF size and accounts for anisotropic seeing. Galaxies that did not satisfy this criterion were classified as unresolved and were excluded from the matched barred–unbarred comparison.

This criterion is intentionally conservative. Previous studies suggest that stellar bars can generally be identified when their characteristic length exceeds roughly ~ 2 –3 times the PSF FWHM (e.g., Aguerri et al. 2009; Liang et al. 2024). However, our quantity $sma_{\max}$ is not a direct measurement of the bar length; rather, it traces the maximum radial extent over which reliable isophotal fitting can be performed. We therefore use $sma_{\max}/FWHM_{\text{eff}} > 4$ as a conservative resolution requirement for identifying bar structure, rather than as a direct translation of the literature bar-length threshold.

Following the ELLIPSE fitting results, our sample consists of 105 barred, 208 unbarred, 42 highly inclined, and 137 unresolved NLS1s. Representative examples of barred, unbarred, and inclined systems are shown in Figure 2.

2.3. Subsample

The barred and unbarred samples exhibit significantly different redshift distributions, reflecting the well-known selection bias against bar detection at higher redshifts. To mitigate this bias, as well as the effects of galaxy

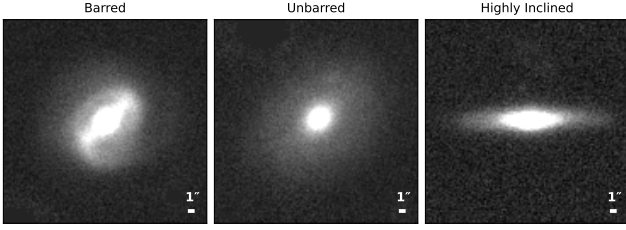


Figure 2. Pan-STARRS cutouts of representative examples of barred, unbarred, and inclined systems.

inclination, we construct a matched subsample by pairing each barred galaxy with the nearest unbarred counterpart in the two-dimensional redshift–inclination parameter space using the *NearestNeighbors* algorithm implemented in *scikit-learn* (Pedregosa et al. 2011). Each barred galaxy is paired with the nearest unbarred galaxy without replacement, resulting in 101 matched pairs. Because the parent sample is drawn from a flux-limited survey, matching in redshift ensures that both subsamples are subject to the same luminosity selection function.

A Kolmogorov–Smirnov (K–S) test confirms that the redshift and inclination distributions of the two subsamples are statistically indistinguishable ($p \approx 0.99$). We further compare the distributions of the number of stacked r -band exposures (N_r) and the FWHM_{eff} to ensure comparable flux depth and spatial resolution. Both K–S and Welch t -test indicate no significant differences between the two subsamples (Figure 3).

We also examine potential projection effects in the bar identification. Since inclinations are derived from the ellipticity of the outer disk, the e_{max} values of unbarred galaxies are expected to follow the apparent ellipticity of a projected disk. The dashed curve in Figure 4 corresponds to the expected relation as mentioned in subsection 2.2. The unbarred galaxies approximately follow this trend, indicating that their measured ellipticity is dominated by disk inclination. In contrast, barred galaxies show a much weaker dependence on inclination and typically exhibit larger ellipticities than expected from projection alone, suggesting that the ellipticity peak primarily traces the bar structure rather than the projected disk.

In summary, the barred and unbarred subsamples have consistent redshifts, inclinations, image quality, and spatial resolutions. This matching minimizes the possibility that observed differences in NLS1 properties are driven by distance or observational biases. We therefore adopt this carefully matched subsample for the subsequent analysis.

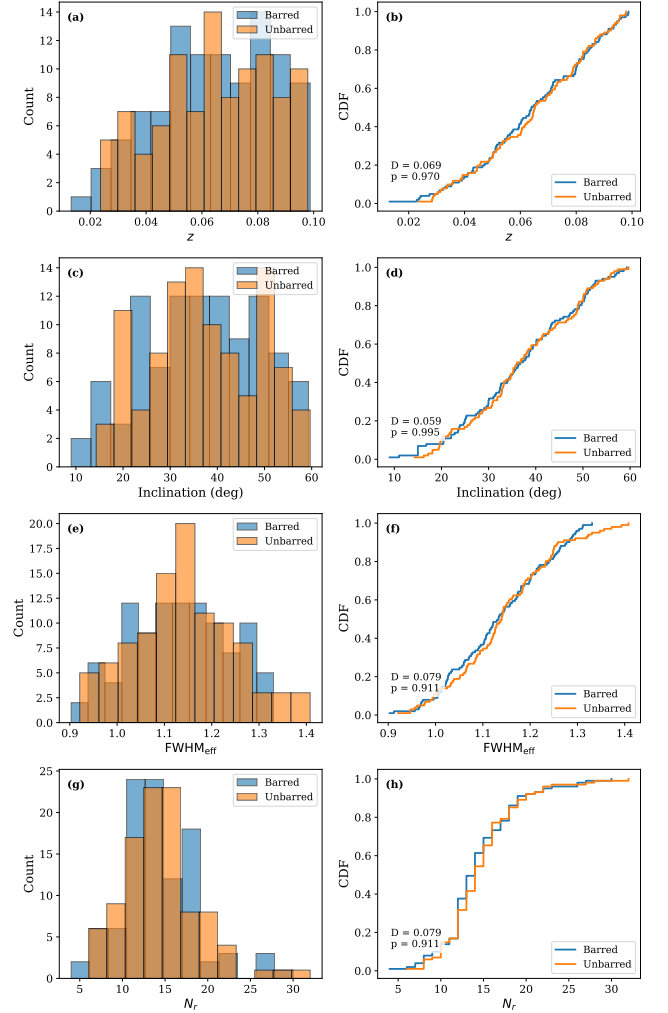


Figure 3. Comparison of the basic properties between the barred and unbarred NLS1 subsamples. The left panels show histograms and the right panels show the corresponding cumulative distribution functions (CDFs). Panels (a) and (b) present the redshift distribution, panels (c) and (d) show the inclination angle, panels (e) and (f) show the effective PSF size FWHM_{eff} , and panels (g) and (h) show the number of stacked r -band images N_r . The Kolmogorov–Smirnov test statistics are indicated in the CDF panels.

3. SPECTROSCOPIC COMPARISON AND STATISTICAL ANALYSIS

We compared catalog-provided spectroscopic data from Paliya et al. (2024) for barred and unbarred NLS1s with K-S test and Welch t -test. All statistical results are listed in Table 1. Significance tests were further verified with Monte Carlo resampling and Bayesian hierarchical modeling (Salvatier et al. 2015).

The Monte Carlo resampling analysis explicitly accounts for measurement uncertainties. For each galaxy, the observed luminosity was perturbed by drawing a

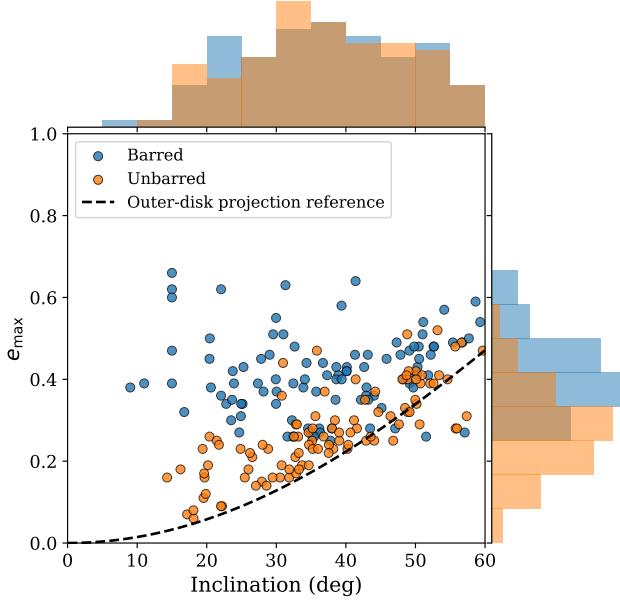


Figure 4. The maximum ellipticity as a function of galaxy inclination for the barred and unbarred NLS1 subsamples. The histograms show the distributions of inclination and ellipticity. The ellipticity peak in unbarred galaxies closely follows the expected disk projection relation, whereas barred galaxies show significantly higher ellipticities and a weaker dependence on inclination.

random value from a Gaussian distribution centered on the measured value with a standard deviation equal to its quoted uncertainty, assuming that the measurement errors are independent and well described by normal distributions. This procedure propagates measurement uncertainties but does not account for potential systematic errors or sample selection effects. For each realization, we recomputed the mean luminosity difference between the barred and unbarred subsamples, as well as the corresponding Welch t -test. This procedure was repeated 10,000 times for each luminosity indicator.

We also modeled the luminosity differences with a simple Bayesian hierarchical model. For each luminosity quantity, the observed value y_i was treated as a noisy measurement of a latent true luminosity θ_i with observational uncertainty σ_i . The latent luminosities were drawn from barred and unbarred parent distributions with group means μ_{bar} and μ_{unbar} and a common intrinsic scatter τ . We then inferred the posterior distribution of $\Delta = \mu_{\text{unbar}} - \mu_{\text{bar}}$ and report the posterior mean of Δ and $P(\Delta > 0)$. The model was implemented with PyMC (Salvatier et al. 2015), using the No-U-Turn Sampler (NUTS; Hoffman & Gelman 2014). We ran four independent chains with 2000 tuning steps and 10,000 posterior draws per chain. This analysis accounts for individual measurement uncertainties and intrinsic pop-

ulation scatter, and provides a probabilistic consistency check on the Monte Carlo results.

3.1. Nuclear Continuum and Broad-Line Emission

In Paliya et al. (2024), the AGN and host-galaxy starlight were decomposed using the BADASS package (Sexton et al. 2021, 2022) to obtain both emission-line and continuum properties. The spectra were shifted to the rest frame, corrected for Galactic extinction, and modeled with a combination of stellar templates, AGN continuum, and emission-line components using a Bayesian Markov Chain Monte Carlo framework (Foreman-Mackey et al. 2013).

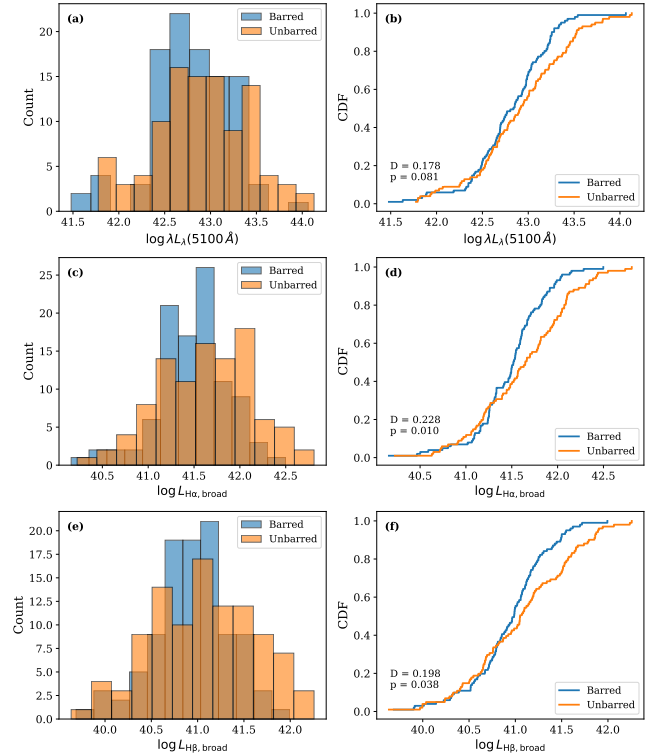


Figure 5. Distributions of the nuclear continuum and broad-line luminosities for the barred and unbarred NLS1 subsamples. The left panels show histograms, while the right panels show the corresponding cumulative distribution functions (CDFs). Panels (a) and (b) present the AGN continuum luminosity $\lambda L_{\lambda}(5100 \text{ \AA})$, panels (c) and (d) show the broad $H\alpha$ luminosity, and panels (e) and (f) show the broad $H\beta$ luminosity. The Kolmogorov-Smirnov test statistics are indicated in the CDF panels.

The monochromatic luminosity at $5100 \text{ \AA}$, denoted as $\lambda L_{\lambda}(5100 \text{ \AA})$, is commonly used as a tracer of the optical AGN continuum (Kaspi et al. 2000; Vestergaard & Peterson 2006), as this spectral region is relatively free from strong emission-line contamination. The unbarred NLS1s show a modestly higher mean

Table 1. Comprehensive Statistical Comparison between Barred and Unbarred NLS1s

Property	Barred Mean $\pm$ Std	Unbarred Mean $\pm$ Std	Sample Size (N_B, N_U)	Stat. Significance p_{KS} / p_{Welch}
(1)	(2)	(3)	(4)	(5)
<i>Basic Properties</i>				
Redshift	0.0644 ± 0.0207	0.0650 ± 0.0199	(101, 101)	0.970 / 0.835
Inclination (deg)	36.66 ± 12.11	37.13 ± 11.50	(101, 101)	0.995 / 0.777
PS1 r -band FWHM (arcsec)	1.134 ± 0.104	1.142 ± 0.107	(101, 101)	0.912 / 0.569
PS1 N_r	14.34 ± 4.43	14.66 ± 4.28	(101, 101)	0.912 / 0.595
<i>Continuum and Broad-Line Luminosities (log erg s⁻¹)</i>				
$\log[\lambda L_\lambda(5100 \text{ \AA})]$	42.80 ± 0.43	42.91 ± 0.52	(101, 101)	0.081 / 0.089
$\log L_{H\alpha, \text{broad}}$	41.49 ± 0.39	41.63 ± 0.51	(101, 101)	0.010 / 0.024
$\log L_{H\beta, \text{broad}}$	40.94 ± 0.42	41.06 ± 0.55	(101, 101)	0.038 / 0.078
<i>Narrow-Line Luminosities (log erg s⁻¹)</i>				
$\log L_{H\alpha, \text{narrow}}$	40.43 ± 0.56	40.62 ± 0.56	(95, 98)	0.023 / 0.015
$\log L_{H\beta, \text{narrow}}$	39.63 ± 0.59	39.87 ± 0.60	(94, 97)	0.0098 / 0.0068
$\log L([\text{O III}] \lambda 5007)$	40.52 ± 0.41	40.69 ± 0.45	(101, 101)	0.0065 / 0.0047
$\log L([\text{N II}] \lambda 6585)$	40.49 ± 0.43	40.56 ± 0.53	(98, 98)	0.105 / 0.343
<i>Line Ratios and Gas Diagnostics</i>				
$N2 \equiv \log([\text{N II}] \lambda 6585/H\alpha)$	0.077 ± 0.318	-0.058 ± 0.288	(89, 92)	0.011 / 0.003
$O3Hb \equiv \log([\text{O III}] \lambda 5007/H\beta)$	0.910 ± 0.547	0.850 ± 0.485	(89, 92)	0.900 / 0.431
$O3N2$	0.833 ± 0.520	0.907 ± 0.511	(89, 92)	0.340 / 0.336
$[S II] \lambda 6716/\lambda 6731$	1.183 ± 0.162	1.164 ± 0.162	(96, 91)	0.322 / 0.436
$n_e([\text{S II}]) \text{ (cm}^{-3}\text{)}$	408 ± 365	497 ± 714	(93, 87)	0.248 / 0.297
<i>Black Hole and Accretion Properties</i>				
$\log M_{BH} (M_\odot)$	6.25 ± 0.39	6.34 ± 0.40	(101, 101)	0.081 / 0.112
$\log \lambda_{Edd}$	-0.625 ± 0.442	-0.598 ± 0.433	(101, 101)	0.970 / 0.668
$\log FWHM_{H\alpha, \text{broad}}$	3.101 ± 0.138	3.125 ± 0.163	(101, 101)	0.820 / 0.252
$\log FWHM_{H\beta, \text{broad}}$	3.107 ± 0.161	3.110 ± 0.139	(101, 101)	0.820 / 0.906
<i>Extinction, Fe II Strength, and Radio Properties</i>				
$\log(H\alpha/H\beta)_{\text{narrow}}$	0.734 ± 0.336	0.677 ± 0.371	(83, 87)	0.253 / 0.292
R_{4570}	0.685 ± 0.327	0.595 ± 0.351	(101, 101)	0.159 / 0.060
$\log R_L$	0.637 ± 0.367	0.426 ± 0.464	(16, 30)	0.079 / 0.099

NOTE—Column (1) lists the galaxy or AGN property being compared. Columns (2) and (3) give the mean and standard deviation for the barred and unbarred NLS1 subsamples, respectively. Column (4) gives the numbers of barred and unbarred objects used in each comparison. Column (5) gives the p -values from the two-sample Kolmogorov–Smirnov test and the independent Welch t -test, respectively. Logarithmic quantities are in base 10.

continuum luminosity than the barred NLS1s, with $\langle \log[\lambda L_\lambda(5100 \text{ \AA})] \rangle = 42.91 \pm 0.52$ for the unbarred subsample and 42.80 ± 0.43 for the barred subsample. However, the difference is only marginal in the two-sample tests, with $p_{KS} = 0.081$ and $p_{Welch} = 0.089$ (Figure 5).

We also assessed the robustness of the 5100 Å continuum offset using Monte Carlo resampling with measurement uncertainties and the Bayesian model. The Monte Carlo analysis gives a positive mean offset of $\Delta = 0.115 \pm 0.011$ dex, where $\Delta = \langle x \rangle_{\text{unbarred}} - \langle x \rangle_{\text{barred}}$, with all realizations yielding $\Delta > 0$. The Bayesian model gives $\Delta = 0.117 \pm 0.067$ dex and $P(\Delta > 0) = 0.961$. These results indicate that the direction of the continuum offset is robust, although its statistical significance remains modest.

We further compared the luminosities of the broad H α and H β components between the barred and unbarred NLS1s (Figure 5). The broad H α luminosity shows a clearer offset: the barred NLS1s have $\langle \log L_{H\alpha, \text{broad}} \rangle = 41.49 \pm 0.39$, while the unbarred NLS1s

have a higher mean value of 41.63 ± 0.51 . This difference is significant in both the K–S and Welch tests, with $p_{KS} = 0.010$ and $p_{Welch} = 0.024$. The broad H β luminosity shows the same direction of offset, with mean values of $\langle \log L_{H\beta, \text{broad}} \rangle = 40.94 \pm 0.42$ for barred NLS1s and 41.06 ± 0.55 for unbarred NLS1s. The K–S test indicates a difference between the two distributions ($p_{KS} = 0.038$), while the Welch test gives a marginal result ($p_{Welch} = 0.078$).

The Monte Carlo and Bayesian analyses support the same overall trend. For broad H α , the Monte Carlo resampling gives $\Delta = 0.145 \pm 0.001$ dex, while the Bayesian model gives $\Delta = 0.145 \pm 0.064$ dex with $P(\Delta > 0) = 0.988$. For broad H β , the corresponding offsets are $\Delta = 0.122 \pm 0.003$ dex from the Monte Carlo analysis and $\Delta = 0.122 \pm 0.069$ dex from the Bayesian model, with $P(\Delta > 0) = 0.963$. Thus, the broad-line results are consistent with the continuum comparison: unbarred NLS1s are modestly but systematically brighter

than barred NLS1s in optical AGN luminosity indicators.

3.2. Narrow-Line Luminosities

We also compared the narrow-line luminosities between the barred and unbarred subsamples. The unbarred NLS1s are systematically brighter in the main narrow recombination lines. For narrow $H\alpha$, the barred subsample has $\langle \log L_{H\alpha, \text{barred}} \rangle = 40.43 \pm 0.56$, while the unbarred subsample has 40.62 ± 0.56 . This difference is significant in both the K-S and Welch tests, with $p_{\text{KS}} = 0.023$ and $p_{\text{Welch}} = 0.015$. For narrow $H\beta$, the barred and unbarred mean luminosities are $\langle \log L_{H\beta, \text{barred}} \rangle = 39.63 \pm 0.59$ and 39.87 ± 0.60 , respectively, with $p_{\text{KS}} = 0.0098$ and $p_{\text{Welch}} = 0.0068$.

The Monte Carlo resampling and the Bayesian model support the same direction of offset. For narrow $H\alpha$, the Monte Carlo analysis gives $\Delta = 0.197 \pm 0.013$ dex, while the Bayesian model gives $\Delta = 0.183 \pm 0.078$ dex with $P(\Delta > 0) = 0.991$. For narrow $H\beta$, the corresponding offsets are $\Delta = 0.236 \pm 0.055$ dex from the Monte Carlo analysis and $\Delta = 0.221 \pm 0.086$ dex from the Bayesian model, with $P(\Delta > 0) = 0.995$. Here $\Delta = \langle x \rangle_{\text{unbarred}} - \langle x \rangle_{\text{barred}}$.

The $[\text{O III}] \lambda 5007$ emission line is commonly used as a tracer of NLR emission associated with AGN photoionization (Kauffmann et al. 2003; Heckman et al. 2005), although it can also be affected by NLR gas content, covering factor, density, and extinction. In the PS1-resolved matched sample, the unbarred NLS1s show higher $[\text{O III}]$ luminosities than the barred NLS1s. The barred subsample has a mean $\log L([\text{O III}] \lambda 5007) = 40.52 \pm 0.41$, while the unbarred subsample has a higher mean value of 40.69 ± 0.45 . The two-sample tests indicate that this difference is significant, with $p_{\text{KS}} = 0.0065$ and $p_{\text{Welch}} = 0.0047$. The Monte Carlo analysis gives a positive offset of $\Delta = 0.172 \pm 0.005$ dex, and the Bayesian model gives $\Delta = 0.171 \pm 0.061$ dex with $P(\Delta > 0) = 0.998$. This result is consistent with the broader trend that barred NLS1s are not enhanced in optical AGN luminosity.

The $[\text{N II}] \lambda 6585$ line is a low-ionization forbidden transition that can arise from both the NLR and other ionized gas components. Unlike $[\text{O III}]$, its strength depends not only on the AGN ionizing radiation, but also on gas-phase metallicity and local ionization conditions. In contrast to the $[\text{O III}]$ result, the $[\text{N II}]$ luminosity does not show a significant barred–unbarred difference. The barred and unbarred mean values are $\log L([\text{N II}] \lambda 6585) = 40.49 \pm 0.43$ and 40.56 ± 0.53 , respectively, with $p_{\text{KS}} = 0.105$ and $p_{\text{Welch}} = 0.343$.

Overall, the narrow-line results show that unbarred NLS1s are brighter in narrow $H\alpha$, narrow $H\beta$, and $[\text{O III}]$

emission, whereas $[\text{N II}]$ luminosity remains statistically comparable between the two subsamples.

3.3. Ionization and Physical Conditions of the Narrow-Line Regions

We examined several narrow-line diagnostic ratios to assess whether barred and unbarred NLS1s differ in the physical conditions of their NLRs. Figure 6 presents the Baldwin–Phillips–Terlevich (BPT) diagram (Baldwin et al. 1981) for the barred and unbarred NLS1 subsamples. Requiring valid measurements for $[\text{O III}]$, $H\beta$, $[\text{N II}]$, and $H\alpha$ leaves 89 barred and 92 unbarred NLS1s. Both populations lie predominantly above the Kewley et al. (2001) demarcation line, indicating that the narrow-line emission is dominated by AGN photoionization rather than star formation.

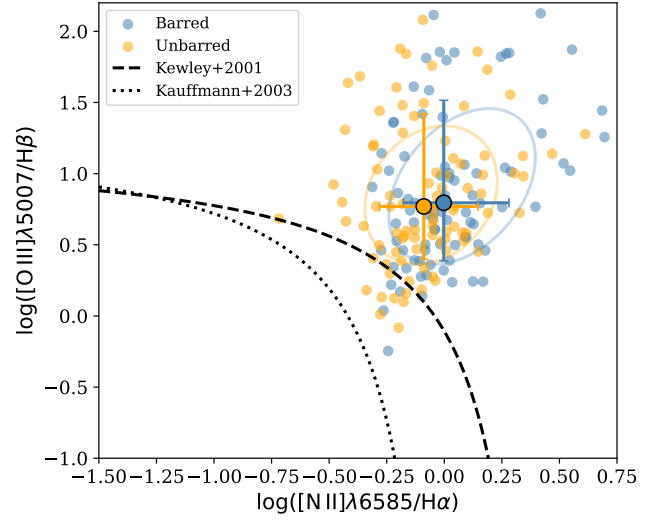


Figure 6. The BPT diagram for barred (blue) and unbarred (orange) NLS1 galaxies. Both populations occupy the AGN region with no obvious separation in $\log([\text{O III}] \lambda 5007 / H\beta)$, although barred systems show a mild shift toward higher $\log([\text{N II}] \lambda 6585 / H\alpha)$. The dashed and dotted curves show the demarcation lines from Kewley et al. (2001) and Kauffmann et al. (2003), respectively.

The ionization-sensitive ratio $\log([\text{O III}] \lambda 5007 / H\beta)$ shows no significant difference between the two subsamples. The barred and unbarred mean values are 0.910 ± 0.547 and 0.850 ± 0.485 , respectively, with $p_{\text{KS}} = 0.900$ and $p_{\text{Welch}} = 0.431$. This indicates that the characteristic NLR ionization state, as traced by $[\text{O III}] / H\beta$, is broadly similar in barred and unbarred systems.

A clearer offset is seen in the N2 index, defined as $\log([\text{N II}] \lambda 6585 / H\alpha)$. The barred subsample has a higher mean N2 value of 0.077 ± 0.318 , compared with -0.058 ± 0.288 for the unbarred subsample. This difference is significant in both the K-S and Welch tests,

with $p_{\text{KS}} = 0.011$ and $p_{\text{Welch}} = 0.003$. Thus, barred NLS1s show relatively stronger low-ionization nitrogen emission with respect to narrow $\text{H}\alpha$.

The O3N2 index, defined here as $\log([\text{O III}] \lambda 5007 / \text{H}\beta) - \log([\text{N II}] \lambda 6585 / \text{H}\alpha)$, compares the relative strength of high-ionization and low-ionization narrow-line emission. The mean O3N2 values are 0.833 ± 0.520 for barred NLS1s and 0.907 ± 0.511 for unbarred NLS1s. Although the unbarred subsample has a slightly higher mean O3N2 value, the difference is not statistically significant, with $p_{\text{KS}} = 0.340$ and $p_{\text{Welch}} = 0.336$. Therefore, the N2 offset is not accompanied by a significant separation in O3N2.

We also used the $[\text{S II}]$ doublet ratio, $[\text{S II}] \lambda 6716 / \lambda 6731$, as a density diagnostic. The barred and unbarred subsamples show statistically consistent $[\text{S II}]$ ratios, with mean values of 1.183 ± 0.162 and 1.164 ± 0.162 , respectively ($p_{\text{KS}} = 0.322$, $p_{\text{Welch}} = 0.436$). Using PyNeb and assuming $T_e = 10^4$ K, the corresponding electron densities are $n_e = 408 \pm 365 \text{ cm}^{-3}$ for barred NLS1s and $497 \pm 714 \text{ cm}^{-3}$ for unbarred NLS1s. These densities are not significantly different ($p_{\text{KS}} = 0.248$, $p_{\text{Welch}} = 0.297$).

Taken together, these diagnostic ratios indicate that barred and unbarred NLS1s have broadly similar NLR ionization states and electron densities. The main exception is the N2 ratio, which is higher in barred systems. However, this difference is not accompanied by a significant enhancement of the $[\text{N II}]$ luminosity itself. As shown in the narrow-line luminosity comparison, $[\text{N II}]$ luminosity remains statistically comparable between the two subsamples, whereas narrow $\text{H}\alpha$ is lower in barred NLS1s. The higher N2 ratio in barred systems therefore reflects the different relative behavior of $[\text{N II}]$ and lower $\text{H}\alpha$, rather than a simple enhancement of $[\text{N II}]$ emission.

3.4. Dependence on Bar Strength

We investigated whether the apparent stellar-bar strength, quantified by e_{max} , correlates with AGN luminosity indicators within the barred subsample. Bar ellipticity is commonly used as a photometric proxy for bar strength (Marinova & Jogee 2007; Aguerri et al. 2009), although it does not directly measure the gravitational bar torque. For the full barred sample ($N = 101$), e_{max} shows a mild but statistically significant anti-correlation with the optical continuum luminosity $\lambda L_{\lambda}(5100 \text{ \AA})$ (Figure 7). The Spearman rank coefficient is $\rho = -0.273$ with $p = 0.0058$, while the Pearson test gives $r = -0.247$ with $p = 0.0128$.

Similar anti-correlations are found for the broad Balmer-line luminosities. For broad $\text{H}\alpha$, the Spear-

man coefficient is $\rho = -0.257$ with $p = 0.0095$, and the Pearson coefficient is $r = -0.226$ with $p = 0.023$. For broad $\text{H}\beta$, the corresponding values are $\rho = -0.271$ with $p = 0.0061$ and $r = -0.260$ with $p = 0.0086$. In contrast, no statistically significant correlation is found between e_{max} and the narrow-line $[\text{O III}] \lambda 5007$ luminosity ($\rho = -0.153$, $p = 0.127$).

To examine possible projection effects, we divided the barred sample into inclination bins of 0° – 20° , 20° – 40° , and 40° – 60° . The lowest-inclination bin contains only eight objects and is therefore too small for a meaningful statistical interpretation. In the intermediate inclination range (20° – 40° , $N = 52$), the anti-correlation between e_{max} and $\lambda L_{\lambda}(5100 \text{ \AA})$ remains significant, with $\rho = -0.313$ and $p = 0.0237$; the Pearson test gives $r = -0.320$ and $p = 0.0209$. The broad-line luminosities show similar negative trends in this bin, although they do not reach formal statistical significance. In the higher-inclination bin (40° – 60° , $N = 41$), the correlations remain negative but are statistically marginal, with $\rho = -0.276$ and $p = 0.080$ for the continuum luminosity.

These results indicate that the apparent bar-strength–continuum anti-correlation is present in the full barred sample and is most clearly detected in systems with moderate inclination. The trend weakens in smaller inclination subsamples and at higher inclination, where projection effects and deprojection uncertainties can increase the scatter in the measured bar ellipticity. Thus, stronger observed bar signatures are not associated with enhanced AGN luminosity. The result is consistent with Silva-Lima et al. (2022).

3.5. Balmer Decrement and Obscuration

To explore whether differential dust attenuation could account for the observed luminosity offsets, we compared both the narrow-line and broad-line Balmer decrements, $\log(F_{\text{H}\alpha}/F_{\text{H}\beta})$, between the barred and unbarred subsamples (Figure 8).

For the broad-line ratio, we treat $\log(F_{\text{H}\alpha}/F_{\text{H}\beta})_{\text{broad}}$ as a broad-line Balmer-decrement or BLR color proxy. Although Dong et al. (2008) showed that the intrinsic broad-line $\text{H}\alpha/\text{H}\beta$ ratio is tightly distributed around 3.06 (with a scatter of ~ 0.03 dex) and largely independent of AGN properties, this intrinsic value differs from, and shows nonzero scatter relative to, the essentially fixed narrow-line Case-B ratio of 2.86 (Osterbrock & Ferland 2006).

For the narrow-line Balmer decrement, the barred subsample has $\langle \log(F_{\text{H}\alpha}/F_{\text{H}\beta})_{\text{narrow}} \rangle = 0.734 \pm 0.336$, while the unbarred subsample has a comparable mean

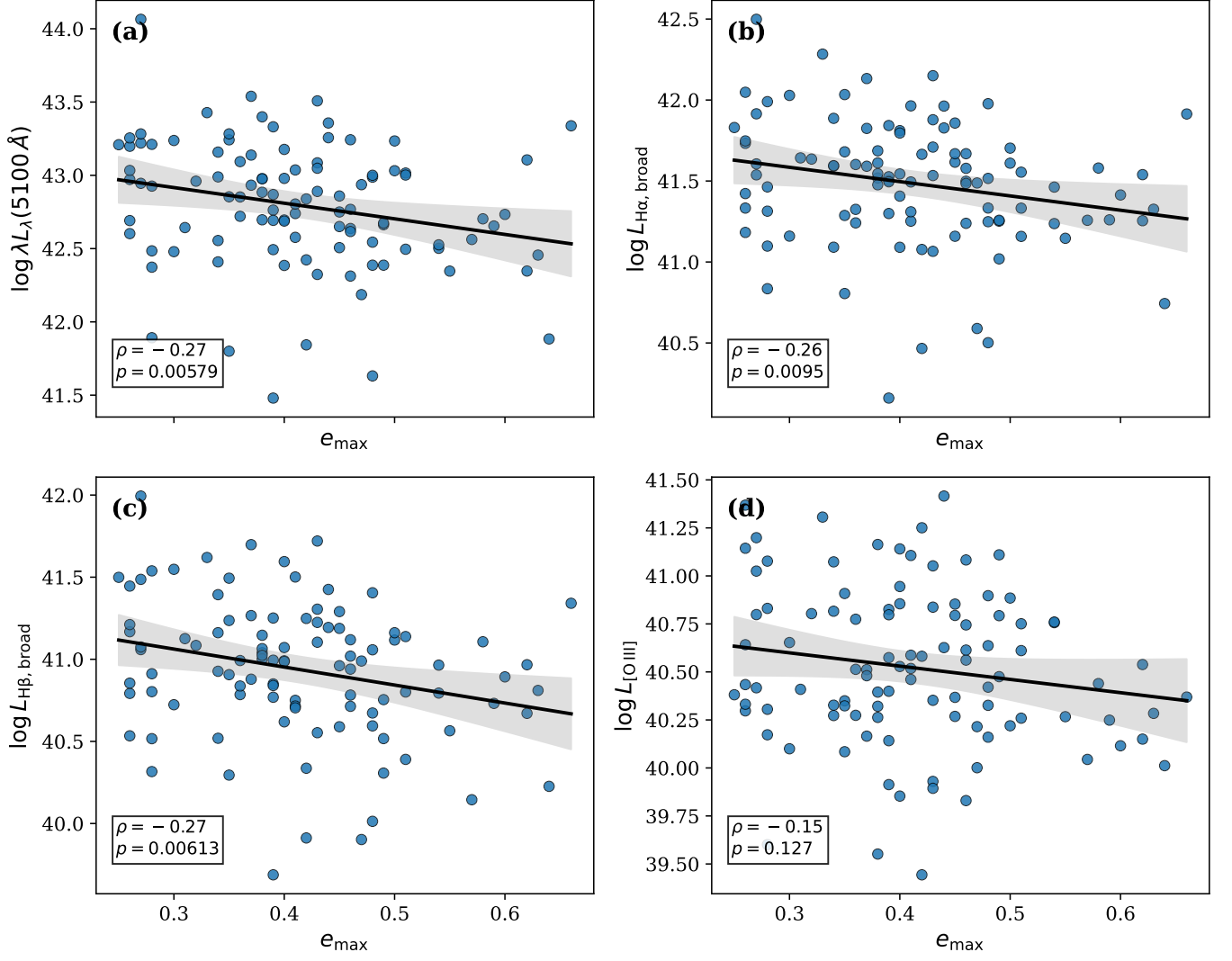


Figure 7. Relation between bar strength ($e_{\max}$) and AGN luminosity indicators for the barred NLS1 subsample: (a) $\lambda L_{\lambda}(5100 \text{ \AA})$, (b) broad H α luminosity, (c) broad H β luminosity, and (d) [O III] luminosity. The black lines show the best-fit linear relations and the shaded regions indicate the 95% confidence intervals.

value of 0.677 ± 0.371 . The difference is not statistically significant, with $p_{\text{KS}} = 0.253$ and $p_{\text{Welch}} = 0.292$.

The broad-line Balmer decrement also shows no significant barred–unbarred difference. The barred and unbarred subsamples have mean values of $\langle \log(F_{\text{H}\alpha}/F_{\text{H}\beta})_{\text{broad}} \rangle = 0.546 \pm 0.123$ and 0.569 ± 0.131 , respectively. The two-sample tests give $p_{\text{KS}} = 0.159$ and $p_{\text{Welch}} = 0.194$.

The similarity of both the narrow-line and broad-line Balmer decrements indicates no evidence for systematically stronger dust attenuation in the barred subsample. Therefore, the systematically lower optical continuum and emission-line luminosities observed in barred NLS1s are unlikely to be primarily caused by enhanced dust extinction in barred systems.

We also examined possible X-ray obscuration using archival X-ray source catalogs. The final optical sample was cross-matched with the Chandra Source Catalog 2.1 (CSC 2.1; Evans et al. 2024) using a search radius of $3''$, and with the HEASARC XMM-Newton Serendipitous Source Catalog DR 15 (5XMM-DR15; Watson et al. 2009; Rosen et al. 2016; Webb et al. 2020) using a search radius of $5''$. The XMMSSC cross-match yields 12 barred and 15 unbarred detections out of 101 objects in each subsample, corresponding to detection fractions of 11.9% and 14.9%, respectively. The CSC 2.1 cross-match yields 18 barred and 13 unbarred detections, corresponding to detection fractions of 17.8% and 12.9%. These detection fractions are not significantly different between the barred and unbarred subsamples.

with Fisher’s exact test ($p_{\text{Fisher}} = 0.680$ for XMMSSC and $p_{\text{Fisher}} = 0.435$ for CSC 2.1).

Among the detected sources, we use the catalog hardness ratios as auxiliary diagnostics of possible X-ray absorption. The XMMSSC hardness ratios show no significant barred–unbarred differences (HR2: $p_{\text{KS}} = 0.990$, $p_{\text{Welch}} = 0.940$; HR3: $p_{\text{KS}} = 0.151$, $p_{\text{Welch}} = 0.285$). Similarly, the CSC 2.1 hardness ratios show no significant differences (HR_{HM}: $p_{\text{KS}} = 0.979$, $p_{\text{Welch}} = 0.856$; HR_{MS}: $p_{\text{KS}} = 0.355$, $p_{\text{Welch}} = 0.200$). Because X-ray absorption tends to harden the observed spectrum, the lack of a significant hardness-ratio difference suggests that the optical luminosity offset is unlikely to be driven primarily by systematically stronger X-ray obscuration in barred systems. However, the X-ray detected subsamples are small, so this comparison is used only as a supplementary check rather than a primary diagnostic.

Taken together, these results indicate that neither optical reddening nor X-ray obscuration differs significantly between barred and unbarred systems. Therefore, attenuation effects are unlikely to drive the observed luminosity offset. However, because the X-ray detected subsamples are small, differences in nuclear obscuration between the two populations cannot be entirely excluded.

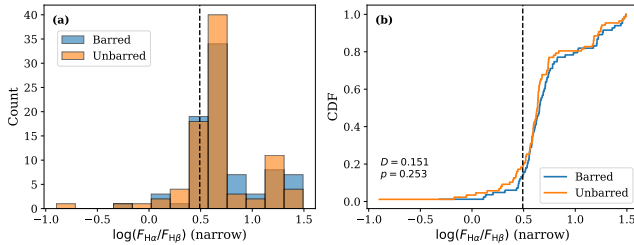


Figure 8. Distribution of the narrow-line Balmer decrement, $\log(F_{\text{H}\alpha}/F_{\text{H}\beta})$, for the barred and unbarred NLS1 subsamples. Panel (a) shows the histogram, and panel (b) shows the cumulative distribution function. The two subsamples are statistically indistinguishable.

3.6. Black Hole Mass and Eddington Ratio

We compared the catalog-based black hole mass (M_{BH}) and Eddington ratio (λ_{Edd}) distributions between the barred and unbarred NLS1 subsamples. The barred galaxies have a mean $\log(M_{\text{BH}}/M_{\odot}) = 6.25 \pm 0.39$, while the unbarred systems have a slightly larger mean value of 6.34 ± 0.40 . However, this difference is not statistically significant, with $p_{\text{KS}} = 0.081$ and $p_{\text{Welch}} = 0.112$.

Similarly, the Eddington ratio distributions are nearly identical for the two populations. The barred subsample has a mean $\log \lambda_{\text{Edd}} = -0.625 \pm 0.442$, compared with

-0.598 ± 0.433 for the unbarred subsample. Both the K–S test ($p_{\text{KS}} = 0.970$) and the Welch test ($p_{\text{Welch}} = 0.668$) indicate no statistically significant difference. These results show that the barred and unbarred subsamples have similar black hole masses and Eddington ratios, despite the modest luminosity offsets seen in the optical continuum and emission-line luminosities.

Because the black hole masses are single-epoch virial estimates that depend on broad-line widths and empirical radius–luminosity relations, we also compare the broad H α and broad H β FWHM distributions (see Table 1). Neither line width shows a significant barred–unbarred difference. Thus, despite the modest luminosity offsets seen in the optical continuum and emission-line luminosities, the two subsamples show no evidence for significantly different virial black hole masses, Eddington ratios, or broad-line kinematics.

3.7. Fe II Strength

Strong Fe II emission, quantified by R_{4570} , is a defining property of NLS1 galaxies (Osterbrock & Pogge 1985). Here R_{4570} is defined as the flux ratio of the Fe II $\lambda 4570$ blend to the broad component of H β . The barred subsample has a mean $R_{4570} = 0.685 \pm 0.327$, while the unbarred subsample has $R_{4570} = 0.595 \pm 0.351$. The difference is not significant in the K–S test ($p_{\text{KS}} = 0.159$), although the Welch test gives a marginal result ($p_{\text{Welch}} = 0.060$). It should be noted that R_{4570} is normalized by the broad H β flux. Therefore, the slightly higher mean R_{4570} in barred NLS1s should not be interpreted directly as evidence for intrinsically stronger Fe II emission; it may partly reflect the lower broad H β luminosity in the barred subsample. Overall, the R_{4570} comparison does not suggest that the main luminosity offsets are driven by a strong systematic difference in BLR emission-line physics.

3.8. Radio Loudness

We examined the radio-loudness parameter, $\log R_{\text{L}}$, for the subsample with available radio information in Paliya et al. (2024). R_{L} is based on the ratio of the rest-frame flux densities at 5 GHz and 4400 Å (Kellermann et al. 1989). In Paliya et al. (2024), the 5 GHz flux density was extrapolated from the 1.4 GHz integrated flux density by assuming a spectral index of $\alpha = 0.5$, where $F_{\nu} \propto \nu^{-\alpha}$.

Radio-loudness measurements are available for 16 barred and 30 unbarred NLS1s. The barred subsample has a slightly higher mean value, $\langle \log R_{\text{L}} \rangle = 0.637 \pm 0.367$, compared with 0.426 ± 0.464 for the unbarred subsample. However, this difference is not statistically significant, with $p_{\text{KS}} = 0.079$ and $p_{\text{Welch}} = 0.099$. Using

the conventional radio-loud threshold $\log R_L > 1$, the radio-loud fractions are nearly identical: 2/16 for barred NLS1s and 4/30 for unbarred NLS1s, with Fisher’s exact test giving $p_{\text{Fisher}} = 1.0$. Therefore, radio-loud objects are unlikely to drive the optical luminosity offsets discussed above, although the radio comparison is limited by the small number of sources with available radio measurements.

4. DISCUSSION

4.1. A Systematic Optical Luminosity Offset

The primary result of this work is that barred NLS1s do not show enhanced optical AGN luminosity relative to their unbarred counterparts. Instead, the unbarred subsample is modestly but consistently brighter in several optical AGN luminosity indicators, including $\lambda L_\lambda(5100 \text{ \AA})$, broad $H\alpha$, broad $H\beta$, narrow $H\alpha$, narrow $H\beta$, and $[\text{O III}] \lambda 5007$, with typical offsets of $\sim 0.1\text{--}0.25$ dex. If the offset were seen only in narrow lines, it could be affected by NLR covering factor, density structure, gas geometry, or the ionizing photon flux intercepted by the NLR gas. However, the same sense of offset is also present in the optical continuum and broad Balmer lines. Since the BLR is much closer to the central engine (e.g., Peterson 1993; Osterbrock & Ferland 2006), and is not governed by large-scale NLR gas geometry, the combined continuum, BLR, and NLR behavior suggests a coherent offset in the overall optical AGN luminosity scale, rather than an effect confined to the NLR.

This luminosity offset is not accompanied by a comparable change in the main narrow-line diagnostic ratios. In particular, the $\log([\text{O III}] \lambda 5007/H\beta)$ ratio is statistically consistent between the barred and unbarred subsamples, with mean values of 0.910 ± 0.547 and 0.850 ± 0.485 , respectively ($p_{\text{KS}} = 0.900$ and $p_{\text{Welch}} = 0.431$). Thus, although both narrow $H\beta$ and $[\text{O III}]$ luminosities are lower in barred NLS1s, their ratio does not show evidence for a significant barred–unbarred difference. The barred and unbarred systems also occupy broadly similar positions in the BPT diagram, and the $[\text{S II}]$ density diagnostic shows no significant difference between the two subsamples. These results suggest that the narrow-line luminosity offset is mainly a normalization difference, rather than a clear change in the characteristic NLR ionization state or electron density.

The luminosity offset is also unlikely to be caused primarily by differential dust attenuation or by other obvious secondary effects. Both the narrow-line and broad-line Balmer decrements show no significant barred–unbarred differences. Similarly, the black hole mass, Eddington ratio, radio-loud fraction, and archival X-ray hardness ratios do not show evidence for a system-

atic difference capable of explaining the optical luminosity offset. Taken together, these results indicate that large-scale stellar bars are not associated with enhanced instantaneous optical AGN luminosity in our matched NLS1 sample.

4.2. Possible Inner Transport Bottlenecks in Barred Systems

Large-scale stellar bars can efficiently redistribute angular momentum and drive gas toward the central kiloparsec. However, this does not guarantee continuous gas delivery to the immediate vicinity of the black hole. From a dynamical perspective, bar strength is itself the outcome of secular angular-momentum redistribution within the galaxy. In isolated barred-galaxy simulations, stronger and longer bars form when more angular momentum is emitted by the inner disk and absorbed by resonant material in the outer disk or spheroidal components; the same exchange also causes the bar pattern speed to decrease with time (Athanasoula 2003; López et al. 2024; Semczuk et al. 2024). Therefore, a stronger large-scale bar signature should be interpreted primarily as evidence for efficient secular redistribution of angular momentum on galactic scales, not as a direct measure of the instantaneous gas inflow rate onto the black hole.

The final transport from circumnuclear scales to sub-100 pc scales requires additional angular momentum loss and can be affected by dynamical barriers such as inner Lindblad resonances, nuclear rings, central mass concentrations, or secondary non-axisymmetric structures (Shlosman et al. 1989; Athanasoula 1992; Combes 2001; Englmaier & Shlosman 2004; Kormendy & Kennicutt 2004; Athanasoula 2013; Combes et al. 2019; Audibert et al. 2019). In this picture, a bar may build up a circumnuclear gas reservoir without necessarily producing an enhancement of instantaneous black hole accretion.

Our results are consistent with this inner-transport bottleneck scenario. Barred NLS1s do not show enhanced optical AGN luminosity relative to unbarred NLS1s, and the apparent bar strength, e_{max} , is mildly anti-correlated with the optical continuum and broad Balmer-line luminosities within the barred subsample. Since e_{max} is an apparent quantity affected by projection and host-galaxy structure (Zou et al. 2014), this trend should not be interpreted as direct evidence that intrinsically stronger bars suppress accretion. Theoretical studies show that intrinsic bar strength depends on the history of angular momentum exchange between the disk, halo, and bulge, as well as on their mass distributions and velocity dispersions (Athanasoula 2003). The observed e_{max} –luminosity trend may therefore reflect a

combination of bar evolutionary stage, host dynamics, and inner fueling efficiency.

The presence of the luminosity offset in the continuum, broad lines, and narrow lines may also provide a timescale clue. The continuum and broad lines trace the more immediate ionizing output, whereas the NLR responds over longer timescales (e.g., Peterson 1993; Benkert et al. 2002). The fact that the offset is also present in narrow $H\alpha$, narrow $H\beta$, and $[O III]$ suggests that the difference is not purely an instantaneous fluctuation confined to the innermost accretion flow. However, the present data cannot determine whether the offset reflects a long-lived secular difference or an episodic duty-cycle difference between barred and unbarred systems.

If the luminosity offset persisted over very long timescales, one might expect a corresponding difference in accumulated black hole growth. The absence of a significant M_{BH} difference suggests that the luminosity offset does not translate into a clearly detectable difference in accumulated black hole growth within the uncertainties of single-epoch virial mass estimates (Vestergaard & Peterson 2006; Shen & Kelly 2010). Larger samples, spatially resolved measurements of circumnuclear gas and star formation, and simulations that follow both bar evolution and black hole fueling will be needed to distinguish these possibilities.

4.3. Limitations

We do not explicitly match the barred and unbarred subsamples in host stellar mass. Reliable stellar masses for Type 1 AGNs require dedicated AGN–host decomposition, and simple broad-band estimates can be affected by AGN continuum, emission-line contamination, and host-galaxy structure (e.g., Kim et al. 2008). In addition, stellar mass may be part of the physical connection between bars and host galaxies, rather than just a matching variable. Large-scale bars are connected to disk structure and secular evolution, so aggressively matching stellar mass could also remove physically relevant differences associated with bar formation. We therefore match the samples in redshift, inclination, image quality, and spatial resolution, which directly control bar detectability and observational selection.

A residual host-mass difference is unlikely to explain the direction of our main result. Since bar fractions generally increase with host stellar mass (Skibba et al. 2012; Gavazzi et al. 2015), any residual mass difference would most naturally make the barred subsample more massive on average. If the optical luminosity offset were driven mainly by host mass, for example through the M_{BH} – M_* relation and comparable Eddington ratios, the barred subsample would then be expected to be the more

luminous one. This is opposite to what we observe. Consistently, the two subsamples show statistically indistinguishable M_{BH} and Eddington-ratio distributions (Section 3.6). Thus, while the lack of direct host-mass matching remains a limitation, a residual host-mass difference is unlikely to account for the lower optical AGN luminosities of barred NLS1s.

Our bar classification relies on isophotal ellipse fitting, which becomes less sensitive when a bright nuclear point source dominates the inner isophotes. Since NLS1s are high-Eddington systems, some genuinely barred galaxies, especially the most AGN-dominated ones, may fail the bar criteria and be assigned to the unbarred subsample. This possible incompleteness would preferentially move AGN-bright barred systems into the unbarred subsample. This effect would dilute the observed barred–unbarred difference. In the limit of severe incompleteness, the two subsamples would become more similar and the luminosity offset would be reduced. Therefore, bar incompleteness could affect the measured amplitude of the offset, but it would not support a scenario in which large-scale bars are associated with enhanced optical AGN luminosity. Within the limits of the present bar classifications, our main conclusion remains that barred NLS1s do not show enhanced present-day optical AGN luminosity relative to their unbarred counterparts.

5. CONCLUSIONS

We have conducted a systematic comparison between barred and unbarred NLS1 galaxies using PS1-resolved matched subsamples controlled in redshift, inclination, and image quality. Our main findings can be summarized as follows:

1. Barred NLS1s do not show enhanced optical AGN luminosity relative to their unbarred counterparts. Instead, the unbarred subsample is modestly but consistently brighter in several optical AGN luminosity indicators, including $\lambda L_\lambda(5100 \text{ \AA})$, broad $H\alpha$, broad $H\beta$, narrow $H\alpha$, narrow $H\beta$, and $[O III] \lambda 5007$. The typical offsets are ~ 0.1 – 0.25 dex.
2. The luminosity offset is not accompanied by a significant difference in the main NLR ionization or density diagnostics. The $\log([O III] \lambda 5007/H\beta)$ ratio, O3N2 index, and $[S II]$ -based electron density are statistically consistent between the barred and unbarred subsamples. The barred systems show a higher N2 ratio, but this is not accompanied by a significant enhancement of the $[N II]$ luminosity itself.

3. The observed luminosity differences are unlikely to be driven primarily by differential dust attenuation, $\log M_{\text{BH}}$, $\log \lambda_{\text{Edd}}$, radio-loud contamination, or X-ray obscuration. Both the narrow-line and broad-line Balmer decrements show no significant barred-unbarred difference. The two subsamples also have statistically similar $\log M_{\text{BH}}$, $\log \lambda_{\text{Edd}}$, radio-loud fractions, and archival X-ray hardness ratios.
4. Within the barred subsample, the apparent bar strength, quantified by e_{max} , shows a mild anti-correlation with optical continuum and broad Balmer-line luminosities. This trend is most clearly detected in systems with moderate inclination. No significant correlation is found between e_{max} and the narrow-line [O III] luminosity.

Taken together, these results suggest that large-scale stellar bars are not associated with enhanced instantaneous optical AGN luminosity in the matched NLS1 sample. A possible explanation is that bars can move gas into the central region of a galaxy, but this gas does not necessarily reach the black hole continuously. Gas may instead remain on circumnuclear scales before smaller-scale processes drive it further inward. In this case, barred systems can contain central gas reservoirs without showing stronger nuclear activity at the time of observation. The modestly lower optical luminosities of barred NLS1s therefore suggest that the bar-AGN connection is not a simple one. Large-scale stellar bars may affect where gas accumulates and when it is delivered inward, rather than directly increasing the instantaneous accretion power of the AGN.

ACKNOWLEDGMENTS

P.-C. Y. is supported by the Taiwan Astronomical Research Alliance (TARA) and Institute of Astronomy and Astrophysics, Academia Sinica (ASIAA), with funding from the National Science and Technology Council of Taiwan (NSTC 115-2740-M-008-001). TARA is committed to advancing astronomy in Taiwan and paving the way for the establishment of a national observatory. The Pan-STARRS1 Surveys (PS1) and the PS1 public science archive have been made possible through contributions by the Institute for Astronomy, the University of Hawaii, the Pan-STARRS Project Office, the Max-Planck Society and its participating institutes, the Max Planck Institute for Astronomy, Heidelberg and the Max Planck Institute for Extraterrestrial Physics, Garching, The Johns Hopkins University, Durham University, the University of Edinburgh, the Queen's University Belfast, the Harvard-Smithsonian Center for Astrophysics, the

Las Cumbres Observatory Global Telescope Network Incorporated, the National Central University of Taiwan, the Space Telescope Science Institute, the National Aeronautics and Space Administration under Grant No. NNX08AR22G issued through the Planetary Science Division of the NASA Science Mission Directorate, the National Science Foundation Grant No. AST-1238877, the University of Maryland, Eotvos Lorand University (ELTE), the Los Alamos National Laboratory, and the Gordon and Betty Moore Foundation.

Funding for the Sloan Digital Sky Survey V has been provided by the Alfred P. Sloan Foundation, the Heising-Simons Foundation, the National Science Foundation, and the Participating Institutions. SDSS acknowledges support and resources from the Center for High-Performance Computing at the University of Utah. SDSS telescopes are located at Apache Point Observatory, funded by the Astrophysical Research Consortium and operated by New Mexico State University, and at Las Campanas Observatory, operated by the Carnegie Institution for Science. The SDSS web site is <https://www.sdss.org>.

SDSS is managed by the Astrophysical Research Consortium for the Participating Institutions of the SDSS Collaboration, including the Carnegie Institution for Science, Chilean National Time Allocation Committee (CNTAC) ratified researchers, Caltech, the Gotham Participation Group, Harvard University, Heidelberg University, The Flatiron Institute, The Johns Hopkins University, L'Ecole polytechnique fédérale de Lausanne (EPFL), Leibniz-Institut für Astrophysik Potsdam (AIP), Max-Planck-Institut für Astronomie (MPIA Heidelberg), Max-Planck-Institut für Extraterrestrische Physik (MPE), Nanjing University, National Astronomical Observatories of China (NAOC), New Mexico State University, The Ohio State University, Pennsylvania State University, Smithsonian Astrophysical Observatory, Space Telescope Science Institute (STScI), the Stellar Astrophysics Participation Group, Universidad Nacional Autónoma de México, University of Arizona, University of Colorado Boulder, University of Illinois at Urbana-Champaign, University of Toronto, University of Utah, University of Virginia, Yale University, and Yunnan University.

This research has made use of the High Energy Astrophysics Science Archive Research Center (HEASARC), a service of the Astrophysics Science Division at NASA/GSFC, the Chandra Source Catalog, provided by the Chandra X-ray Center (CXC) as part of the Chandra X-ray Observatory mission, and the XMM-Newton 5XMM-DR15 Source Catalogue compiled by the insti-

tutes of the XMM-Newton Survey Science Centre selected by ESA.

The author acknowledges the use of AI-based tools during the preparation of this manuscript. ChatGPT (OpenAI) and Claude (Anthropic) assisted with language editing and the development of analysis code. All AI-assisted code was reviewed, tested, and verified by the author. These tools were not used independently to generate results, interpretations, or conclusions. All

scientific content and conclusions presented in this work remain the sole responsibility of the author.

Software: Python, Astropy (Astropy Collaboration et al. 2013, 2018, 2022), Astroquery (Ginsburg et al. 2019), IRAF (Tody 1986, 1993), Matplotlib (Hunter 2007), NumPy (Harris et al. 2020), pandas (McKinney 2010), PyMC (Salvatier et al. 2015), PyNeb (Luridiana et al. 2015), SciPy (Virtanen et al. 2020), scikit-learn (Pedregosa et al. 2011)

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